

languages that was easy for people to pick up quickly. It is open source and is free to use even for commercial applications. It's used as a scripting language, and programmers can easily produce readable and functional code in a very short period of time.

Features of Python

- Supports procedure-oriented as well as object-oriented programming (OOP). Python has a very powerful but simplistic way of implementing OOP.
- Good readability, with clear and simple syntax.
- Portable, therefore it can be interpreted in various operating systems, including UNIX-based systems, various Microsoft Windows versions, Mac OS, MS DOS, etc.
- The source code is open for users to modify and reuse.
- Easy to learn compared to other such languages.
- It can be embedded in C and C++.
- It's a high-level language that does not need compilation to the binary.

4. Perl

Perl is still under active development even after 27 years. According to the Perl website, it is a stable, mature, powerful and portable language. It is highly efficient and feature rich. Perl 5 runs on over 100 platforms, which vary from portables to mainframes. It is suitable for both rapid prototyping as well as the large-scale development of various projects. Perl 5 and Perl 6 are parts of the Perl family of languages. However, Perl 6 is a separate language that has a different development team working on it. Hence, it comes as a dynamically typed, interpreted language which is comparable with both PHP and Python. It is ideal for processing and producing text data. It is highly portable and widely supported.

Features of Perl

- Has simple syntax, so it's easy to learn for novices as well.
- It is cross-platform.
- Is versatile and has a very comprehensive library of modules.
- Has a very powerful as well as flexible object-oriented programming syntax.
- Supports multiple platforms including Linux, Microsoft DOS, Windows, Apple Mac, and some mobile platforms as well.

How to select a programming language

When we think of choosing our first programming language, several points pass through our minds, and the many options available add to the confusion. There are several factors that can help us select the most optimal programming language.

1. *Performance and efficiency:* This is one parameter that all of us look at while evaluating any programming language. We then choose the one that most efficiently performs a specific task in the least time. In reality,

no specific language is fast. It's the efficiency of the compiler and interpreter that makes it fast. So we should choose accordingly.

2. *Capability to address your specific requirements:* We should choose the one that best addresses our specific problem. We cannot select a language just because it's popular or is highly efficient. It's of no use if it doesn't work to solve our problem.
3. *Ease of learning, understanding and time taken for developing code:* A programming language that takes less time for a coder to understand and learn, enabling it to be implemented for a specific application, is better than one that is complex. Also, if it's easy to understand, it will take less time to code and at the same time, it will be easy for peers to review and modify the code.
4. *Elasticity and portability:* It's always good to opt for languages that can easily be modified in order to support some new feature or functionality, without making too many changes. It is also advantageous if a programming language supports multiple platforms, middleware, databases and system management facilities or if it can support these with very few adjustments.
5. *Security:* A programming language should include specific security measures in order to protect its code from malicious usage.
6. *Popularity and support:* If the language we select is popular, then it becomes easier for us to find reference material. Also, the chances of finding its library files are higher, compared to less popular languages. It will be good if proper support is available as this saves a lot of time.

Open source vs licensed programming languages

The benefits

- One of the most important reasons for preferring open source programming languages is that their source code can be customised efficiently to fit our needs and requirements, whereas in the case of a licensed programming language, if the source code is made available by the owner, it can be customised only to the permissible extent.
- Open source programming languages are subject to peer review constantly, so the bugs present in the program code can be easily found and fixed. Users from different programming backgrounds and countries can collaborate, unlike in the case of commercially developed programs where only the developers of the original piece of code can actually change it.
- Open source programming languages also allow the translation of the source code from one language to another, whereas in the case of licensed programming languages, the developers of the source code may not allow that.
- Open source programming languages are free to try out before actual implementation, unlike licensed ones, where we have trial versions that are valid for a few days only.
- Open source programming languages are more secure overall.